CB^2-Reg: design-matrix beta-binomial regression for quantitative and covariate-adjusted CRISPR pooled screens

Hyun-Hwan Jeong

July 23, 2026

Abstract

CB^2 was developed for a common but narrow pooled-screen question: does guide abundance differ between two experimental groups? Modern screens increasingly measure trajectories, doses, ordered phenotypes, and multifactor designs that cannot be reduced to one pairwise contrast without discarding information. We introduce CB^2-Reg, a design-matrix extension that preserves the original beta-binomial sampling logic while replacing the two-group comparison with regression coefficients and prespecified linear contrasts. We show that the original CB^2 statistic admits an algebraically exact generalized-least-squares representation on its group-summary scale. This is a legacy-compatibility identity, not strict nesting of the default estimator. Under a common-dispersion model, local alternatives, and increasing numbers of independent libraries, the logit-scale regression statistic is first-order equivalent to the original statistic. Guide counts remain the response; sample phenotypes and nuisance variables enter through a logit mean model; and a Student t reference uses sample-level residual degrees of freedom rather than sequencing reads. The implementation supports continuous and categorical predictors, batch and donor adjustment, interactions, contrasts, empirical-null scaling with prespecified controls, parallel screening, and RcppArmadillo linear algebra.

Four complementary benchmarks define what this regression extension gains and where it should not be used. In simulation, a read-level binomial test is severely anti-conservative, whereas beta-binomial inference restores uncertainty to the sample level. In the longitudinal HT-29 screen, CB^2-Reg achieves the highest recall at 90% precision while remaining comparable with continuous-time MAGeCK-MLE and Chronos in global discrimination. In a primary-T-cell ordered-bin FACS stress test, control-calibrated four-bin CB^2-Reg recovers 22 of 26 experimentally concordant IL2RA regulators versus 17 for all-bin MAGeCK-MLE. Within the selected 33-gene follow-up panel, it has F1 0.863 and Matthews correlation 0.398, compared with 0.723 and 0.070 for MAGeCK-MLE, while restoring the non-targeting-guide $p < 0.05$ frequency from 13.3% to 4.9%. The purpose-built Waterbear result remains the best calibrated modeling choice for correlated FACS partitions and recovers 24 of 26 controls. Endpoint and copy-number analyses further show that no normalization or count model wins automatically in every design. CB^2-Reg is therefore best viewed as a design-matrix beta-binomial extension of the original sampling principle: a fast, transparent general-purpose method for independent libraries with quantitative or adjusted sample designs, accompanied by diagnostics that reveal when a specialist likelihood is required.

1 The problem CB^2-Reg solves

For guide g in sequenced sample i, let K_{gi} be its read count and let N_i be the total number of mapped guide reads in that sample. The observed guide proportion is $Y_{gi} = K_{gi}/N_i$. A sample can also have a continuous measurement d_i, such as drug dose, time, or a quantitative sample phenotype.

Statistically, K_{gi} is the response and d_i is a predictor, even when d_i is called the experimental "outcome" in the biological description.

The scientific question becomes whether guide abundance follows the phenotype after accounting for the experimental design. A single coefficient can represent a dose or time trend; several coefficients can represent donors, batches, treatments, or interactions; and a contrast can test any prespecified one-dimensional comparison. This avoids turning an ordered experiment into arbitrary "high" and "low" groups and makes the adjustment set explicit.

This scope distinction is essential. CB^2-Reg models a phenotype attached to each independently sequenced library. It does not recover one continuous phenotype per cell when guide identity and phenotype are not observed jointly. FACS bins can provide an ordered approximation, but bins produced from the same donor are correlated partitions and require a joint-bin model for fully generative inference.

2 From CB^2 to CB^2-Reg

CB^2 adapts the beta-binomial weighted test of Baggerly et al. [1] to CRISPR pooled screens [2]. Its central insight is worth preserving: sequencing depth measures a guide proportion precisely, but it does not turn reads into independent biological replicates. The original method applies this insight to a two-group contrast. CB^2-Reg retains the same depth-aware, extra-binomial variance principle and changes the mean model from two group means to a design matrix. "Retains" refers to the stochastic hierarchy and the sample-level information ceiling; it does not mean that the default logit fit reproduces the original finite-sample statistic. The legacy representation and asymptotic relationship are established below.

Table 1: What changes—and what does not—from the original CB^2 method to CB^2-Reg.

Feature	Original CB^2	CB^2-Reg
Primary question	Difference between two groups	Trend, adjusted association, interaction, or contrast
Sample design	Reference and comparison labels	Arbitrary full-rank R model matrix
Mean model	Two unrelated group proportions	$\operatorname{logit}(\mu_i) = \mathbf{x}_i^\top \boldsymbol{\beta}$
Effect	Difference or fold change between groups	Conditional log-odds coefficient or linear contrast
Dispersion	Separate beta-binomial weighting within each group	One guide-specific ρ shared across the fitted design
Estimator	Weighted group proportions	Feasible logistic IRLS
Variance principle	Binomial sequencing plus between-library beta-binomial heterogeneity	
Reference df	Group-variance df formula	Residual $m - q$
Adjustment	Pairwise stratification or preprocessing	Batch, donor, dose, time, interactions, splines, and other covariates
Implementation	Existing CB^2 workflow	Additive `bbreg()`, `bb_contrast()`, and `bb_screen()` APIs with RcppArmadillo kernels

The regression bridge was described by Baggerly et al. [3] for sequencing count data with multiple groups and continuous covariates. The method combines logistic regression with extra-binomial variation and retains a coefficient-based t test. Here we state that construction in CRISPR-screen notation and use the beta-binomial, library-size-dependent variance of Williams [4]. An identity-link regression of raw proportions would allow fitted values outside $(0, 1)$, while an unweighted correlation would ignore unequal library sizes. The logit and beta-binomial weights solve those two problems directly.

3 Model

The following model is fitted separately for every guide, so the guide index is suppressed. Conditional on a latent sample-specific abundance P_i,

$$K_i \mid P_i \sim \text{Binomial}(N_i, P_i), \tag{1}$$

$$P_i \sim \text{Beta}(\mu_i \kappa, (1-\mu_i)\kappa). \tag{2}$$

The mean is linked to a row vector of sample covariates $\mathbf{x}_i^\top$ by

$$\text{logit}(\mu_i) = \log\left(\frac{\mu_i}{1-\mu_i}\right) = \mathbf{x}_i^\top \boldsymbol{\beta}. \tag{3}$$

The design can contain an intercept, a continuous phenotype, batch indicators, interactions, spline terms, and other prespecified adjustment variables.

It is useful to express the beta precision κ as an intraclass correlation

$$\rho = \frac{1}{\kappa+1}, \qquad 0 \le \rho < 1. \tag{4}$$

Marginally,

$$\text{E}(K_i) = N_i \mu_i, \tag{5}$$

$$\text{Var}(K_i) = N_i \mu_i (1-\mu_i)\{1 + (N_i - 1)\rho\}, \tag{6}$$

$$\text{Var}(Y_i) = \mu_i(1-\mu_i)\left\{\frac{1-\rho}{N_i} + \rho\right\}. \tag{7}$$

Equation (7) cleanly separates sequencing uncertainty, which decreases with N_i, from between-sample heterogeneity, which does not vanish when sequencing depth becomes arbitrarily large.

3.1 Interpretation of the continuous coefficient

Suppose the model contains a continuous phenotype d_i with coefficient β_d. Then $\exp(\beta_d)$ is the multiplicative change in the odds of observing that guide for a one-unit increase in d_i, conditional on the other covariates. Standardizing d_i before fitting makes $\exp(\beta_d)$ the odds ratio per sample standard deviation and makes guide effects directly comparable. For rare guides, odds and proportions are similar, so β_d is also approximately a log abundance ratio per unit of d_i.

4 Estimation by feasible IRLS

Let X be the $m \times q$ full-rank design matrix. Given current values of $\boldsymbol{\beta}$ and ρ, define

$$\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}, \qquad \mu_i = \frac{\exp(\eta_i)}{1+\exp(\eta_i)}, \tag{8}$$

$$z_i = \eta_i + \frac{Y_i - \mu_i}{\mu_i(1-\mu_i)}, \qquad w_i = \frac{N_i \mu_i (1-\mu_i)}{1 + (N_i - 1)\rho}. \tag{9}$$

The coefficient update is the weighted least-squares step

$$\widehat{\boldsymbol{\beta}}_{\text{new}} = (X^\top W X)^{-1} X^\top W \mathbf{z}, \qquad W = \operatorname{diag}(w_1, \ldots, w_m). \tag{10}$$

This is weighted least squares on the logistic working response, not on the raw proportions.

For a fixed fitted mean, ρ is estimated from the Pearson equation

$$\sum_{i=1}^{m} \frac{(K_i - N_i \widehat{\mu}_i)^2}{N_i \widehat{\mu}_i (1 - \widehat{\mu}_i)\{1 + (N_i - 1)\widehat{\rho}\}} = m - q. \tag{11}$$

If the left side at $\rho = 0$ is no larger than $m - q$, the estimate is truncated at zero rather than claiming negative overdispersion. Equations (9)–(11) are alternated to convergence.

The weight has an important limiting behavior:

$$\lim_{N_i \to \infty} w_i = \frac{\mu_i(1 - \mu_i)}{\rho} \quad \text{when } \rho > 0. \tag{12}$$

Therefore a deeply sequenced sample cannot acquire unlimited leverage. Once sequencing uncertainty is small, biological heterogeneity sets the information ceiling.

5 T-based inference

At convergence, the model-based covariance estimator is

$$\widehat{\operatorname{Cov}}(\widehat{\boldsymbol{\beta}}) = (X^\top \widehat{W} X)^{-1}. \tag{13}$$

The Pearson equation scales the residual variation to approximately one. If the dispersion estimate reaches its numerical upper boundary, the implementation additionally multiplies Equation (13) by the remaining Pearson scale.

Equation (13) is a plug-in, model-based covariance: it conditions on the estimated $\widehat{\rho}$ as though it were fixed. It does not propagate uncertainty from estimating a separate dispersion for every guide. Referring the coefficient to t_{m-q} gives a heavier-tailed small-sample reference than a normal approximation, but it is not a formal correction for dispersion-estimation uncertainty. With few independent libraries, calibration must therefore be checked by simulation, phenotype permutation, or prespecified negative controls.

For coefficient β_j, the statistic

$$t_j = \frac{\widehat{\beta}_j - \beta_{j,0}}{\sqrt{\widehat{\operatorname{Cov}}(\widehat{\boldsymbol{\beta}})_{jj}}} \tag{14}$$

is compared with a Student t_{m-q} distribution. The degrees of freedom are driven by the number of independent sequenced samples, not by the total read count. This is the crucial reason that a library with millions of reads does not create artificial certainty.

More generally, for a prespecified contrast vector $\mathbf{c}$,

$$t_{\mathbf{c}} = \frac{\mathbf{c}^\top \widehat{\boldsymbol{\beta}} - \delta_0}{\sqrt{\mathbf{c}^\top \widehat{\operatorname{Cov}}(\widehat{\boldsymbol{\beta}}) \mathbf{c}}} \tag{15}$$

uses the same $m - q$ degrees of freedom. This covers pairwise group contrasts, average slopes in interaction models, and other one-degree-of-freedom hypotheses. Guide-wise two-sided p-values are adjusted using the Benjamini–Hochberg procedure [5].

5.1 Legacy representation and local-equivalence result

We now separate an algebraic compatibility result from the substantive estimating-equation connection. The first result reproduces CB^2 after its weighted group summaries have already been computed; because a saturated two-cell GLS can represent any pair of independent group summaries, this result should not be interpreted as proof that the implemented `bbreg()` estimator strictly nests the original estimator. For the original test, let

$$\widehat{p}_A = \sum_{i\in A} a_i Y_i, \qquad \widehat{p}_B = \sum_{i\in B} b_i Y_i, \qquad \sum_{i\in A} a_i = \sum_{i\in B} b_i = 1,$$

where the depth- and dispersion-dependent weights a_i, b_i are those estimated by CB^2. Let its estimated group variances be $V_A > 0$ and $V_B > 0$. The original guide statistic and reference degrees of freedom are

$$T_{\mathrm{CB2}} = \frac{\widehat{p}_B - \widehat{p}_A}{\sqrt{V_A + V_B}}, \tag{16}$$

$$\nu_{\mathrm{CB2}} = \frac{(V_A + V_B)^2}{V_A^2/(n_A - 1) + V_B^2/(n_B - 1)}. \tag{17}$$

Lemma 1 (Legacy CB^2 representation)**.** *Let X have row $(1, 0)$ for samples in group A and row $(1, 1)$ for samples in group B. Define the working precision matrix*

$$\Omega = \mathrm{diag}\left(\left\{\frac{a_i}{V_A} : i \in A\right\}, \left\{\frac{b_i}{V_B} : i \in B\right\}\right).$$

Then the identity-link generalized least-squares estimator satisfies

$$\widehat{\boldsymbol{\beta}}_{\mathrm{GLS}} = (X^\top \Omega X)^{-1} X^\top \Omega \mathbf{Y} = \begin{pmatrix} \widehat{p}_A \\ \widehat{p}_B - \widehat{p}_A \end{pmatrix},$$

and

$$(X^\top \Omega X)^{-1} = \begin{pmatrix} V_A & -V_A \\ -V_A & V_A + V_B \end{pmatrix}.$$

Consequently, the GLS t statistic for the slope is exactly T_{CB2}.

Proof. Because the weights within each group sum to one, their groupwise total working precisions are

$$\sum_{i\in A} \frac{a_i}{V_A} = \frac{1}{V_A}, \qquad \sum_{i\in B} \frac{b_i}{V_B} = \frac{1}{V_B}.$$

Hence

$$X^\top \Omega X = \begin{pmatrix} V_A^{-1} + V_B^{-1} & V_B^{-1} \\ V_B^{-1} & V_B^{-1} \end{pmatrix}, \quad X^\top \Omega \mathbf{Y} = \begin{pmatrix} \widehat{p}_A/V_A + \widehat{p}_B/V_B \\ \widehat{p}_B/V_B \end{pmatrix}.$$

Direct inversion gives the displayed covariance matrix and multiplication gives the displayed coefficient vector. Its slope standard error is $\sqrt{V_A + V_B}$, proving the result. □

Corollary 1 (Legacy-compatibility identity)**.** *If the slope statistic in Lemma 1 is referred to $t_{\nu_{\mathrm{CB2}}}$ using Equation (17), its two-sided p-value is identical to the original CB^2 p-value. Thus identity-link GLS algebra reproduces the original two-group inference on its group-summary scale. This corollary is a compatibility check, not a claim of finite-sample equality with the default logit `bbreg()` fit.*

The default implementation uses the logit link because it respects the parameter space for arbitrary designs. It is therefore important to prove a second, weaker relationship rather than claim finite-sample identity.

Proposition 1 (First-order equivalence of the logit contrast)**.** *Suppose the two groups are independent, their weighted proportion estimators are consistent, their groupwise and pooled precision estimators converge to a common* κ*, and under the null or a sequence of local alternatives* $p_A, p_B \to p \in (0,1)$ *with* $\widehat{p}_B - \widehat{p}_A = O_p(n^{-1/2})$*. Let* $h(p) = \operatorname{logit}(p)$ *and define the delta-method statistic*

$$T_{\text{logit}} = \frac{h(\widehat{p}_B) - h(\widehat{p}_A)}{\sqrt{h'(\widehat{p}_A)^2 V_A + h'(\widehat{p}_B)^2 V_B}}.$$

Then

$$T_{\text{logit}} - T_{\text{CB2}} \xrightarrow{p} 0.$$

Under a correctly specified common-dispersion beta-binomial model, the binary-design IRLS Wald statistic has the same first-order expansion. If $n_A, n_B \to \infty$ *through increasing numbers of independent biological libraries, both its residual degrees of freedom and Equation* (17) *diverge, so their reference distributions also converge to the same standard normal limit. Holding* n_A, n_B *fixed while only increasing read depths* N_i *is not this asymptotic regime.*

Proof. The mean-value theorem and consistency give

$$h(\widehat{p}_B) - h(\widehat{p}_A) = h'(p)(\widehat{p}_B - \widehat{p}_A) + o_p(n^{-1/2}),$$

where $h'(p) = \{p(1-p)\}^{-1}$. Continuity of h' gives

$$\sqrt{h'(\widehat{p}_A)^2 V_A + h'(\widehat{p}_B)^2 V_B} = h'(p)\sqrt{V_A + V_B}\{1 + o_p(1)\}.$$

The common positive factor $h'(p)$ cancels, yielding $T_{\text{logit}} = T_{\text{CB2}} + o_p(1)$. It remains to connect this transformed cell-mean statistic to IRLS. Write $\rho = 1/(\kappa+1)$. Within a group having mean μ_g, the beta-binomial quasi-score for its logit cell mean is proportional to

$$\sum_{i \in g} \frac{N_i}{\kappa + N_i}(Y_i - \mu_g).$$

The original CB^2 weight is

$$\left(\frac{1}{\kappa} + \frac{1}{N_i}\right)^{-1} = \frac{\kappa N_i}{\kappa + N_i}.$$

The factor κ disappears on normalization, so for fixed common dispersion the two methods use identical within-group relative weights. Under the stated common-dispersion model, the separately estimated legacy precisions and the IRLS precision converge to the same κ; the resulting binary-design cell means therefore have the same first-order expansion. Finally, both t distributions approach the standard normal as their degrees of freedom diverge. □

These results establish two deliberately separate claims. The original statistic has an *algebraically exact legacy representation* after its group summaries are computed, while the implemented logit statistic is *locally asymptotically equivalent* in the unadjusted binary design under the stated common-dispersion assumptions. Strict finite-sample nesting of the implemented estimator is not claimed. In finite samples, `bbreg(~group)` need not equal `measure_sgrna_stats()`: the effect scale, dispersion pooling, working weights, and degrees of freedom differ. What regression adds is an estimable coefficient for continuous predictors, adjustment variables, and interactions; what it changes must be assessed by calibration and benchmark data rather than hidden behind the word "generalization." The executable example `examples/cb2_generalization_proof.R` verifies Lemma 1 to machine precision and illustrates the convergence in Proposition 1. In its deterministic guide, original CB^2 and the GLS contrast both give effect 9.059929×10^{-4}, standard error 1.113105×10^{-4}, $t = 8.139330$, $\nu = 5.647255$, and two-sided $p = 2.512928 \times 10^{-4}$; the maximum numerical discrepancy is 1.78×10^{-15}. In the link-scale illustration, shrinking the two-group proportion difference from 8×10^{-4} to 2.5×10^{-5} reduces the absolute gap between raw and logit statistics from 0.0583 to 0.000151.

6 Calibration: why the beta-binomial layer matters

The script `examples/simulation.R` simulates 600 guides targeting 200 genes in 27 independent samples. The samples span nine standardized phenotype values in each of three batches, with library sizes between 40,000 and 120,000. Each gene has three guides; 160 genes have a zero phenotype coefficient, 20 have coefficient 0.9, and 20 have coefficient -0.9. Counts are drawn from a beta-binomial model with $\rho = 0.0015$. All methods adjust for batch. Guide-level beta-binomial and naive-binomial results are summarized by the within-gene median effect and directional Stouffer evidence. Official MAGeCK-MLE 0.5.9.5 receives the identical continuous design matrix and its Wald inference is reported.

Table 2: Results from the simulation with random seed 20260723. Type-I error and power use unadjusted $p < 0.05$; discoveries and empirical FDR use Benjamini–Hochberg adjusted $p < 0.05$.

Method	Type I	Power	Discoveries	Empirical FDR	RMSE	95% coverage
Naive binomial z	0.844	1.000	174	0.770	0.158	0.185
Beta-binomial t	0.081	1.000	46	0.130	0.156	0.935
Official MAGeCK-MLE Wald	0.038	0.975	41	0.049	0.140	0.970

This simulation is a calibration experiment rather than a contest designed to make CB^2-Reg win. The binomial-only test estimates the slopes reasonably well, but its standard errors treat reads as independent biological replicates. Its null type-I error is therefore 0.844 rather than 0.05, and most reported discoveries are false. The beta-binomial model estimates almost the same effects while inflating uncertainty to reflect between-sample variation. Its type-I error, empirical FDR, and confidence-interval coverage improve substantially while power remains 1.00. In this run, official MAGeCK-MLE is best calibrated: its type-I error is 0.038, empirical FDR is 0.049, and coverage is 0.970, with power 0.975. Therefore the simulation does not support claiming that beta-binomial regression is uniformly better. Monte Carlo variation is expected because only 160 null genes are used in this single reproducible run.

One simulated guide with true coefficient 0.9 gives

$$\widehat{\beta}_d = 1.159, \qquad \mathrm{SE} = 0.267, \qquad t_{23} = 4.341, \qquad p = 2.41 \times 10^{-4},$$

with $\widehat{\rho} = 5.58 \times 10^{-4}$. MAGeCK-MLE gives a gene effect of 0.851 with Wald standard error 0.184 and $p = 3.81 \times 10^{-6}$. The diagnostic plot in Figure 1 shows the null calibration and this fitted dose response.

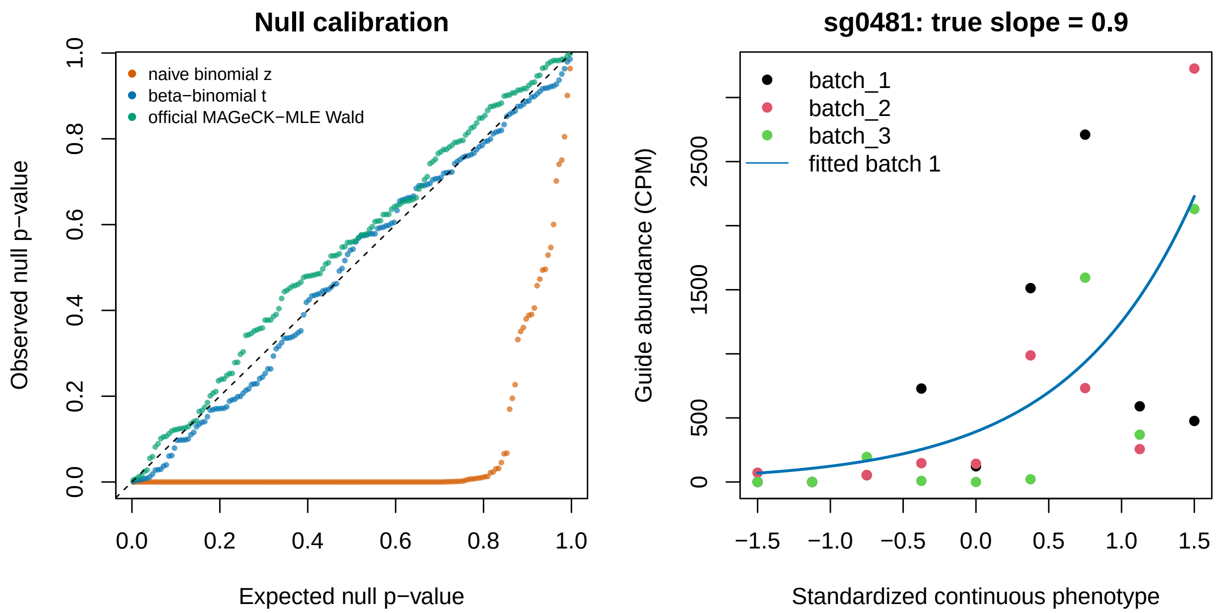

Figure 1: Left: observed versus expected gene-level null p-values for the naive binomial, beta-binomial, and official MAGeCK-MLE fits. Right: observed guide abundance and the fitted batch-1 beta-binomial dose response for one truly associated guide.

7 Backward compatibility: GSE70038

GSE70038 contains 64,747 guides measured in 16 libraries from two glioblastoma stem-cell lines and two neural stem-cell lines, with duplicate initial and terminal samples [6]. This binary terminal-versus-initial setting is a special case of the regression model, not a continuous-phenotype benchmark. It is nevertheless useful because it provides published biology and permits an exact design-matched comparison with MAGeCK-MLE.

We followed the design-matrix rules in Table 5 and Box 3 of the MAGeCKFlute protocol [7]. The first column is one for every library; each initial library has zeros elsewhere; and each terminal library has one in exactly one of four cell-line columns. The same 16×5 matrix was supplied to beta-binomial regression and the official MAGeCK-MLE 0.5.9.5 executable. MAGeCK used median normalization and its Wald inference. For the beta-binomial result, guide effects were summarized by their within-gene median, and directional guide p-values were combined by Stouffer's method. This aggregation is an explicit comparison device, not a proposed replacement for MAGeCK's gene model.

The "effect rank correlation" reported below is calculated in this analysis, not copied from GEO or the MAGeCKFlute paper. Separately for each terminal coefficient, we pair all genes available from both methods. The beta-binomial effect is the within-gene median guide coefficient, and the MAGeCK effect is its official MLE beta. We assign average ranks to these two effect vectors and take their Pearson correlation, which is exactly Spearman's rank correlation. The 18,077 paired effects and their ranks for every coefficient are written to `results/gse70038/effect_concordance_pairs.csv.gz`.

The comparison is not circular: beta-binomial regression conditions on each library total, whereas MAGeCK uses a normalized negative-binomial count model with guide-efficiency estimation [8]. Despite those differences, all four effect correlations exceed 0.88. These correlations measure agreement of the inferred gene ordering; they are not likelihood, calibration, or predictive-fit statistics

Table 3: Effect concordance between beta-binomial regression and official MAGeCK-MLE on GSE70038. Jaccard is the overlap divided by the union of the 200 most depleted genes from each method.

Terminal coefficient	Spearman effect correlation	Top-200 Jaccard
GSC0131	0.928	0.550
GSC0827	0.888	0.533
NSCCB660	0.905	0.544
NSCU5	0.915	0.562

and therefore cannot identify a better-fitting method. In GSC0131, the published convergent dependency PKMYT1 has effect -0.987 and FDR 1.87×10^{-8} under beta-binomial regression, versus beta -0.786 and Wald FDR 4.53×10^{-8} under MAGeCK-MLE. Both also recover HEATR1, TFAP2C, and WEE1 depletion. Figure 2 shows the complete GSC0131 comparison; the reproducible PDF contains one page per terminal coefficient.

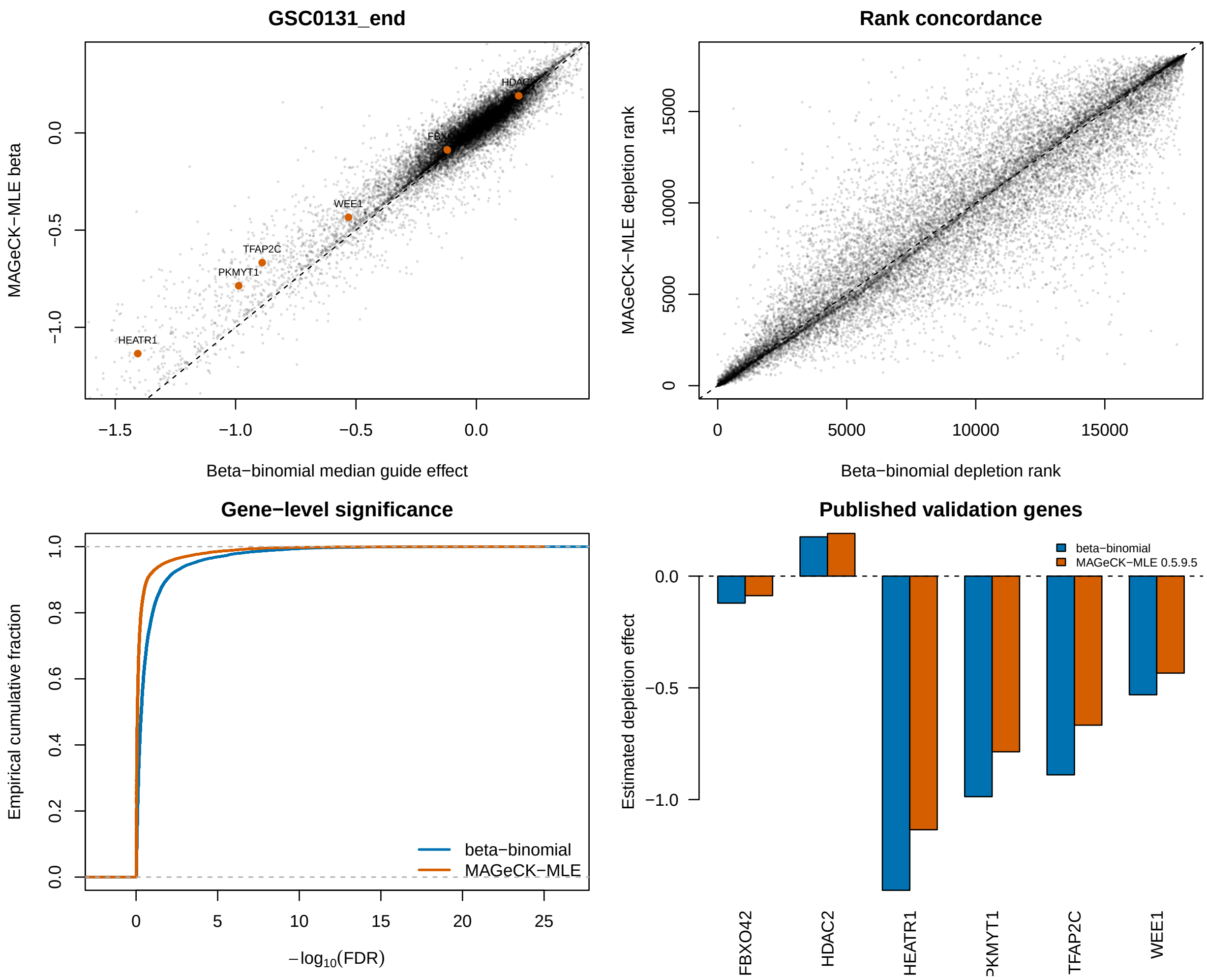

Figure 2: GSE70038 GSC0131 head-to-head. Panels compare gene effects, depletion ranks, FDR distributions, and published validation genes. Orange points in the effect panel mark the prespecified genes PKMYT1, HEATR1, TFAP2C, FBXO42, HDAC2, and WEE1.

8 Primary use case: longitudinal HT-29 screening

The preceding datasets do not exercise the main reason for this extension: a continuous predictor observed at more than two values. We therefore added the HT-29 time course of Tzelepis et al. [9], the longitudinal dataset used to evaluate Chronos [10]. It contains a pDNA measurement and HT-29 guide counts at days 7, 10, 13, 16, 19, 22, and 25. The three sequencing columns at each late day were summed before analysis. This distinction is important for the requested t inference: three assays of one harvested population increase read depth but are not three independent biological units and therefore must not add residual degrees of freedom. Following the Chronos preprocessing, guides with fewer than 30 pDNA reads were removed, leaving 86,882 guides targeting 17,995 genes.

The beta-binomial and official MAGeCK-MLE 0.5.9.5 fits received the same eight-column count matrix and the same numeric design, with an intercept and $d_i/25$. This is a direct continuous-time comparison. We also downloaded the all-seven-time-point Chronos joint result and the deposited per-endpoint MAGeCK and BAGEL2 results from the Chronos Figshare record [11]. As in the Chronos paper, the latter two were combined by taking each gene's median effect over the seven separate endpoint runs.

For an external biological target, core essential genes are positives and HT-29 genes with DepMap 20Q2 expression below 0.5 are negatives, matching the deposited Chronos analysis. All methods were restricted to the same 1,080 essential and 5,774 unexpressed genes. Normalized null-median difference (NNMD) is the essential-gene median minus the unexpressed-gene mean, divided by mean absolute deviation of the unexpressed effects; more negative values indicate stronger separation. "False positives" counts unexpressed genes in the lowest 15% of all evaluated effects, following the deposited Chronos analysis.

Table 4: Longitudinal HT-29 control-gene benchmark. PR AUC is the area under the precision–recall curve; Recall_{90} is the maximum recall at at least 90% precision. More negative NNMD and fewer unexpressed false positives are better.

Method	AUROC	PR AUC	NNMD	Recall_{90}	$\text{FP}_{15\%}$
Beta-binomial time	0.9786	0.9494	−12.35	0.9037	74
Official MAGeCK time	0.9789	0.9534	−13.14	0.8972	76
Chronos joint	0.9747	0.9356	−18.88	0.8991	77
Published MAGeCK average	0.9742	0.9351	−10.62	0.8602	100
Published BAGEL2 average	0.9725	0.9295	−7.28	0.8148	133

Table 4 gives a more useful answer than declaring one universal winner. Official continuous-time MAGeCK has the highest AUROC and PR AUC, although its AUROC advantage over beta-binomial is only 0.0004. Beta-binomial regression has the highest high-precision recall and two fewer unexpressed false positives. Chronos has by far the strongest distributional separation by NNMD, consistent with its mechanistic treatment of knockout efficiency and saturation. The beta-binomial and official continuous-time MAGeCK effects have Spearman correlation 0.926; correlations with Chronos joint and the deposited MAGeCK average are 0.815 and 0.812. These are again effect agreement statistics, not comparable likelihoods: the methods estimate differently scaled quantities and do not expose a common predictive likelihood in the deposited outputs.

8.1 What the longitudinal screen says about CNV correction

We extracted the HT-29 profile (ACH-000552) from DepMap Public 20Q2 [12] and repeated the same copy-number audit used for A375. Before correction, effect–copy-number Spearman correlation among unexpressed genes is already small: −0.072 for beta-binomial, −0.075 for official continuous-time MAGeCK, and −0.048 for deposited Chronos joint effects. Applying MAGeCK 0.5.9.5's official piecewise normalizer makes the association larger rather than smaller: −0.145 for beta-binomial and −0.149 for MAGeCK. Thus the A375 result does not generalize automatically.

This failure mode is informative. The piecewise correction estimates one global effect–CNV trend across genes in one cell line and applies it after model fitting. When the nuisance association is weak and biological dependencies are not exchangeable across copy-number regions, the fitted trend can have the wrong practical target and over-correct null genes. Chronos joint is shown without CNV correction because the published Chronos procedure requires multiple cell lines to identify common essential genes and was explicitly not applied to this single-line experiment [10]. The DepMap profile was measured separately from the original screen, so cell-line drift is an additional limitation; the result supports diagnosing correction rather than applying it by default.

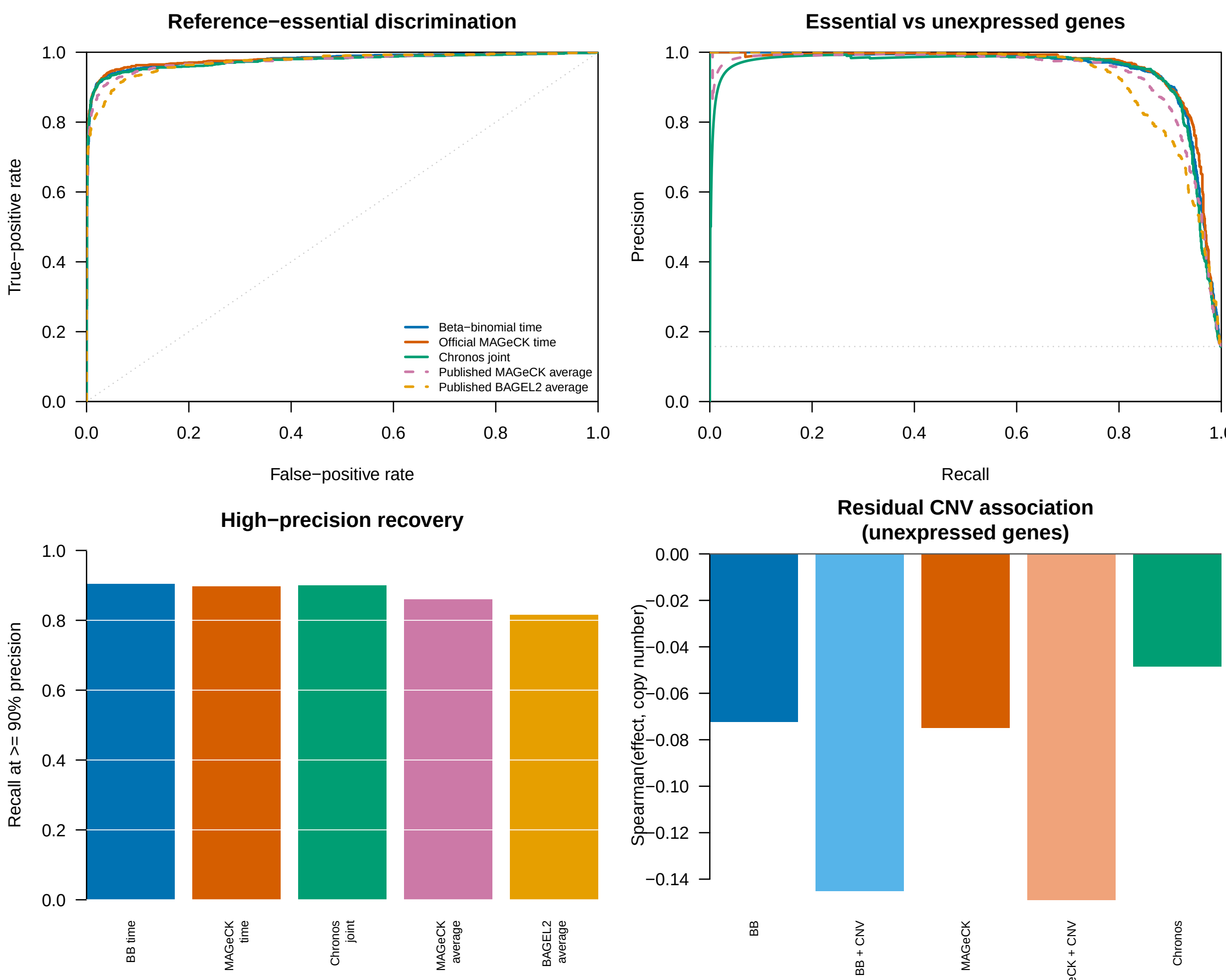

Figure 3: Longitudinal HT-29 benchmark. The upper panels compare control-gene ROC and precision–recall curves. The lower-left panel shows recall at at least 90% precision. The lower-right panel shows residual effect–copy-number association among unexpressed genes before and after official MAGeCK piecewise correction; the more negative corrected bars demonstrate over-correction in this single-cell-line screen.

9 Stress test and model boundary: ordered-bin FACS

The Waterbear study supplies a complementary continuous-phenotype benchmark with unusually useful validation [13]. GSE242880 contains a pooled CRISPR screen of IL2RA protein abundance in primary human T cells. Its low-coverage, high-MOI arm has four ordered FACS bins for each of three donors, 6,000 guides targeting 1,350 genes or negative controls, and 26 regulators whose direction was confirmed by individual knockout and flow cytometry. Because these are primary donor cells rather than a cancer cell line, this benchmark is not driven by tumor copy-number artifacts.

We assigned the four bins latent marker locations equal to the conditional means of standard-normal intervals with the prespecified bin masses $(0.2, 0.3, 0.3, 0.2)$. Beta-binomial regression used all 12 libraries with a marker slope and donor indicators. Official MAGeCK-MLE received the same model space; its marker score was affinely mapped to zero–one because the MAGeCK initializer requires a reference row with all non-intercept entries zero. We also fitted two outer-bin ablations and ran the paper's native `mageck test` comparison of Q1 with Q4.

This design exposes a limitation of applying independent-library regression to sorted data. The four bins from one donor partition one cell pool and are therefore negatively correlated, while the beta-binomial fit treats their margins separately. Among 593 non-targeting guides, the raw all-bin beta-binomial $p < 0.05$ frequency is 0.133. We therefore added the prespecified negative-control calibration implemented by `bb_calibrate_controls()`: the empirical 95th percentile of absolute control t statistics is matched to the t_8 two-sided 0.05 cutoff, with scales below one prohibited. The estimated scale is 1.357; after calibration, the non-targeting frequency is 0.049. Estimates and the t reference are retained, while raw and calibrated inferential columns are both saved.

Table 5: GSE242880 low-coverage, high-MOI IL2RA FACS benchmark. Recovery uses each method's reported 0.10 discovery threshold and the independently validated direction. CB^2-Reg and MAGeCK use adjusted p-values; Waterbear uses posterior inclusion probability above 0.90. The non-targeting column is guide-level. Waterbear and MAUDE values are reported by the study and were not rerun.

Method	Discoveries	Validated recovery	NTC $p < 0.05$
CB^2-Reg, four bins + controls	49	22/26	0.049
CB^2-Reg, four bins raw	127	23/26	0.133
Official MAGeCK-MLE, four bins	72	17/26	—
CB^2-Reg, outer bins	34	18/26	0.064
Official MAGeCK test, outer bins	32	18/26	—
Waterbear, four bins (reported)	79	24/26	—
MAUDE, four bins (reported)	406	25/26	—

The calibrated four-bin CB^2-Reg result recovers four more validated regulators than all-bin MAGeCK-MLE while making 23 fewer calls. It is two regulators behind the reported Waterbear result and three behind MAUDE. The raw 23/26 CB^2-Reg result is not the primary claim because its negative controls reveal inflation. MAUDE's 25/26 recovery is also not evidence of superior specificity: it calls 406 of 1,350 genes, and the Waterbear study found poor MAUDE calibration in simulations [13, 14]. Across all 1,350 genes, uncalibrated CB^2-Reg and MAGeCK-MLE all-bin effects have Spearman correlation 0.917, so the recovery difference is not caused by globally incompatible directions. All 26 validation effects have the expected sign under each independently rerun method.

The follow-up table contains more information than the 26-positive subset. Thirty-three candidates were tested individually: 26 had a concordant IL2RA effect and seven did not validate. Restricting evaluation to these measured labels avoids the incorrect assumption that every other screen gene is a false positive. Table 6 reports threshold and ranking metrics for the five methods rerun here.

Table 6: Classification in the selected 33-gene experimental follow-up panel (26 concordant positives and seven non-validating candidates). This panel is small and was selected from a previous screen, so the metrics are supporting evidence rather than unbiased genome-wide performance estimates.

Method	F1	MCC	Balanced accuracy	AUROC	Avg. precision
CB^2-Reg, four bins + controls	0.863	0.398	0.709	0.808	0.945
CB^2-Reg, four bins raw	0.852	0.194	0.585	0.819	0.950
Official MAGeCK-MLE, four bins	0.723	0.070	0.541	0.626	0.872
CB^2-Reg, outer bins	0.783	0.340	0.703	0.714	0.913
Official MAGeCK test, outer bins	0.766	0.224	0.632	0.703	0.906

The raw four-bin model has the strongest rank metrics, which indicates that the continuous trend is extracting useful signal. Control calibration changes the decision boundary rather than the estimated effects and produces the best F1, Matthews correlation, and balanced accuracy in the validation panel. That combination is the central evidence for CB^2-Reg here: continuous-bin ranking gains are retained while explicit null guides prevent the extra calls from being mistaken for confidence.

This benchmark is stronger than a rank-correlation-only comparison because it contains directionally validated biology and hundreds of explicit null guides. It is not an unbiased precision experiment: the 26 genes were chosen for follow-up from an earlier screen, and unvalidated genes cannot simply be called false positives. Nor does it overturn Waterbear's modeling advantage. Waterbear represents the joint multinomial bin structure, uncertain cutoffs, guide-to-gene hierarchy, and sparse effects; CB^2-Reg supplies a much faster transparent trend test but needs negative controls here to diagnose and scale inference under a violated independence assumption. The deposited paper reports 24/26 for Waterbear and 17/26 for MAGeCK; the independent current `mageck test` rerun gives 18/26, a one-gene implementation-level difference.

These results separate three meanings of "better." First, ranking quality asks whether validated regulators move toward the top; the raw four-bin model performs strongly by AUROC and average precision. Second, decision quality asks whether a prespecified threshold balances recovery and false calls; the control-calibrated model leads the rerun methods by F1 and Matthews correlation. Third, likelihood fidelity asks whether the sampling dependence is represented; Waterbear wins that design question because it models all bins from a donor jointly. CB^2-Reg is compelling here because it does well on the first two questions with a simple general-purpose model and also provides the control diagnostic that exposes the third.

The published Waterbear and MAUDE aggregates cannot be assigned validation- panel F1 or Matthews correlation without their gene-level calls for all seven non-validating candidates. Reporting 24/26 or 25/26 as if it were complete accuracy would therefore mix sensitivity with an unknown precision. We retain those values as positive-control recovery references and restrict classification metrics to the methods rerun from complete outputs.

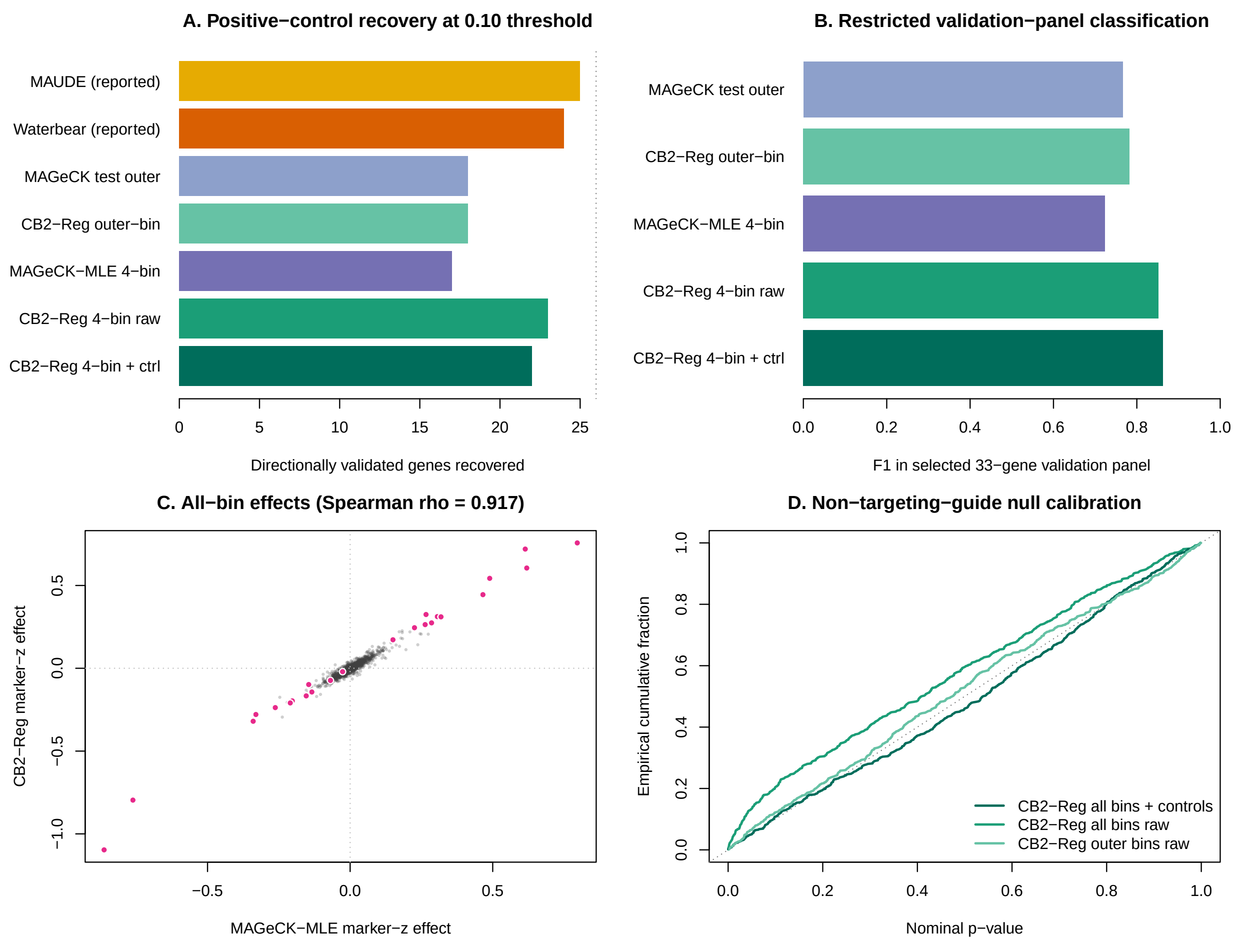

Figure 4: Ordered-bin IL2RA FACS benchmark. Top: positive-control recovery for all methods and F1 in the selected 33-gene panel for methods rerun here. Bottom left: all-bin CB^2-Reg and MAGeCK-MLE effects, with validated genes highlighted. Bottom right: non-targeting-guide empirical null curves, showing the correction of raw all-bin inflation.

10 Endpoint benchmark and copy-number audit

To ask which method ranks biological dependencies better, we used the Brunello A375 negative-selection screen of Sanson et al. [15], bundled with CB^2. This is an ordinary endpoint screen rather than a continuous-phenotype experiment, but it supplies an external target unavailable in GSE70038: reference essential genes should rank ahead of reference nonessential genes. The beta-binomial score combines guide evidence by directional Stouffer aggregation; the MAGeCK score uses the official MLE beta and Wald p-value. Both are oriented so larger scores mean stronger depletion.

10.1 Copy-number-aware analysis

Copy number is a gene-level property of A375, not a sample-level covariate: for a given guide it is constant across the plasmid and three terminal libraries. Adding it directly to the four-row sample design would therefore be non-identifiable with the endpoint effect. Instead, we downloaded the A375 gene-level profile (ACH-000219) from DepMap Public 19Q3 [16, 17] and used MAGeCK 0.5.9.5's official `-cnv-norm` piecewise correction. For an apples-to-apples comparison, the same MAGeCK piecewise normalizer was applied to the beta-binomial within-gene median effects. It adjusts effect sizes after model fitting; it does not refit either count likelihood. The common CNV-complete evaluation set contains 1,506 essential and 869 nonessential genes.

Table 7: CNV-aware Sanson A375 benchmark. The final column is Spearman correlation between effect and copy number among reference nonessential genes; values nearer zero indicate less residual CNV-associated depletion. F1 uses nominal FDR 0.05 and a negative effect.

Method	AUROC	Avg. precision	F1	CNV correlation
Beta-binomial regression	0.9598	0.9786	0.8531	−0.185
Beta-binomial + CNV	0.9533	0.9762	0.8531	−0.042
Official MAGeCK-MLE	0.9598	0.9795	0.8277	−0.207
Official MAGeCK-MLE + CNV	0.9501	0.9762	0.8277	−0.006

CNV correction substantially removes the intended nuisance association: −0.185 to −0.042 for beta-binomial effects and −0.207 to −0.006 for MAGeCK. It does not improve essential-gene discrimination in this screen. AUROC decreases by 0.0066 and 0.0097, respectively, and the 0.05-FDR calls are unchanged. This is not a contradiction: bias removal and gold-standard classification are different targets, and amplified genes can include both dependencies and nondependencies.

After CNV correction, beta-binomial versus MAGeCK global ranking remains indistinguishable: the AUROC difference is 0.00316 (paired stratified-bootstrap 95% CI −0.00265 to 0.00924), and the average precision difference is effectively zero (95% CI −0.00349 to 0.00295). At nominal FDR 0.05, beta-binomial retains 4.18 percentage points more recall (95% CI 2.66 to 5.71) and an F1 advantage of 0.0253 (95% CI 0.0148 to 0.0362).

10.2 Why beta-binomial does not dominate MAGeCK

The A375 screen is favorable to MAGeCK: it is a binary endpoint experiment, MAGeCK's native use case, with one plasmid baseline and three terminal libraries. A guide-wise beta-binomial fit then has only two residual degrees of freedom, making its t tail deliberately conservative. MAGeCK instead borrows information across the screen, estimates guide efficiency, and uses asymptotic Wald inference. These features explain its advantage at very small FDR thresholds; the intended advantage of using a genuinely continuous predictor is not exercised by this dataset.

MAGeCK is nevertheless not an oracle. Its negative-binomial mean–variance model, median normalization, guide-efficiency model, and normal-reference Wald tests are assumptions rather than guarantees [8, 18]. They can be sensitive to global composition changes, atypical guide behavior, or limited biological replication. Moreover, MAGeCK 0.5.9.5 performs CNV normalization after its permutation p-values and FDR values are calculated. Consequently, CNV correction can remove effect–CNV association while leaving the significance calls unchanged, exactly as observed here. Conversely, the present beta-binomial prototype estimates dispersion guide by guide and uses a transparent but ad-hoc Stouffer gene aggregation. A fair path to more consistent gains is empirical-Bayes dispersion moderation plus a native gene-level hierarchy, not changing the benchmark or weakening MAGeCK.

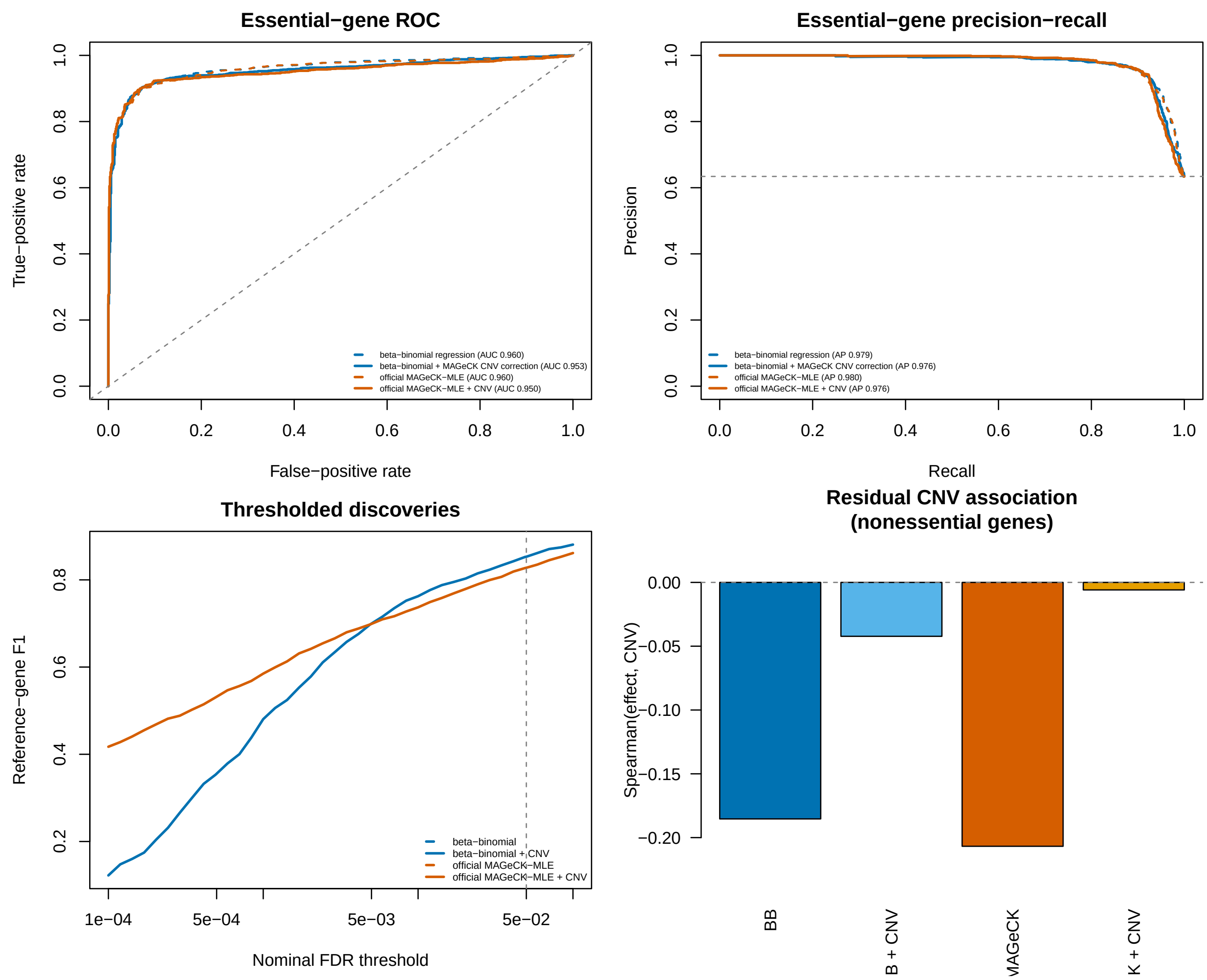

Figure 5: CNV-aware essential-gene discrimination in the Sanson A375 screen. ROC (upper left), precision–recall (upper right), F1 across nominal FDR thresholds (lower left), and residual effect–CNV correlation among nonessential genes (lower right). Curves compare uncorrected (dashed) and CNV-corrected (solid) beta-binomial and official MAGeCK-MLE results.

11 R usage

The standalone beta-binomial implementation uses base R. The same additive API is integrated into the cloned CB^2 package; there its weighted IRLS cross-products and solve are implemented with RcppArmadillo. Existing CB^2 two-group functions remain unchanged: CB^2-Reg is an additive analysis path rather than a breaking replacement.

```
source("R/bbreg.R")

# One guide: K contains guide counts; N contains sample library sizes.
fit <- bbreg(
  count   = K,
  total   = N,
  formula = ~ scale(dose) + batch,
  data    = sample_data
)
summary(fit)

# A named contrast; unspecified coefficient weights are zero.
bb_contrast(fit, c("scale(dose)" = 1))

# A complete guide-by-sample screen.
result <- bb_screen(
  counts  = count_matrix,
  totals  = N,
  data    = sample_data,
  formula = ~ scale(dose) + batch,
  term    = "scale(dose)",
  gene    = guide_annotation$gene,
  ncores  = 4
)

# Optional empirical-null scaling when prespecified negative controls exist.
result <- bb_calibrate_controls(
  result,
  control = guide_annotation$gene == "Non-Targeting Control"
)
result[order(result$fdr), ]
```

Column order in the count matrix must match row order in `sample_data`. The phenotype should be defined before looking at guide results, and all independent biological replicates should remain separate. For the negative-binomial benchmark, the project calls the official `mageck mle` executable with the same design matrix rather than embedding a partial MAGeCK reimplementation in CB^2. On the development machine, the installed CB^2 package analyzes 10,000 guides in 4.73 seconds serially and 1.36 seconds with four forked workers. The longitudinal analysis fits 86,882 filtered guides across eight aggregated time points in 19.4 seconds with four workers (4,481 guides per second); 86,840 fits reached the strict convergence tolerance, and all fits returned finite time coefficients.

12 Choosing between CB^2 and CB^2-Reg

The methods answer related but different questions. Original CB^2 remains the direct choice when the biological estimand is one prespecified difference between two independent conditions. CB^2-Reg should be preferred when the design itself carries information that a pairwise split would discard: three or more ordered levels, a dose or time axis, donor or batch adjustment, an interaction, or a named contrast from a multivariable model.

Table 8: Practical method choice by experimental design.

Design	Primary method	Reason
Two independent groups	CB^2 or MAGeCK	The estimand is already a pairwise contrast
Independent libraries across dose, time, or ordered phenotype	CB^2-Reg	Uses the full quantitative axis with sample-level t inference
Multivariable independent samples	CB^2-Reg	Adjusts batch, donor, and interactions in one explicit model matrix
FACS bins partitioning the same donor pool	Waterbear	Models the joint bin probabilities and within-donor dependence
Multi-line viability dynamics	Chronos, with CB^2-Reg as a transparent trend analysis	Mechanistic growth and knockout timing may matter
Cancer screen with suspected CNV artifact	Diagnose before correction	Post-fit CNV normalization can help, do nothing, or over-correct

This decision rule is also the selling point of the extension. It does not ask users to abandon a validated two-group workflow. It opens the same beta-binomial framework to the large class of pooled screens whose scientific question is a coefficient rather than a group label.

13 Assumptions, diagnostics, and limitations

1. **Independent samples.** The t_{m-q} reference distribution assumes independent sequenced libraries. Repeated measures, matched samples, or multiple libraries from one biological unit require a subject-level correlation model or a blocked permutation. FACS bins from one donor are a compositional partition, not four independent biological replicates; the GSE242880 control calibration is a diagnostic sensitivity analysis, whereas a purpose-built joint model such as Waterbear represents that structure directly.

2. **Enough sample-level degrees of freedom.** Read depth cannot replace biological replication. With very few samples, guide-wise $\widehat{\rho}$ is noisy and coefficient tests may be poorly calibrated. The model-based covariance treats $\widehat{\rho}$ as a fixed plug-in value, and the t_{m-q} reference does not formally propagate its estimation uncertainty. The local-equivalence result likewise requires increasing numbers of independent libraries; deeper sequencing at fixed replication is not sufficient. The optional `bb_calibrate_controls()` procedure additionally requires all control guides to share one residual degree of freedom, as they do when every guide uses the same complete design. Guides fitted with heterogeneous designs require stratified or redesigned calibration. Dispersion shrinkage across guides or permutation of the phenotype should be evaluated before treating this prototype as a production method.

3. **Mean-model form.** A single slope assumes a linear relationship on the logit scale. Residual plots should be checked. A spline or a prespecified nonlinear term is preferable when the

dose response is curved. This matters directly in viability time courses: incomplete knockout can produce finite depletion saturation, which Chronos models mechanistically but a single beta-binomial slope does not.

4. **Sparse and separated guides.** Guides with all zero counts or with counts confined to one end of the phenotype range can cause boundary fits. Low-count filtering should be chosen without reference to the tested phenotype. A penalized or likelihood-ratio procedure is a useful future extension for severe separation.

5. **Compositional counts.** Guide counts share a fixed library total. The marginal beta-binomial model handles depth and guide-specific overdispersion but does not remove global composition shifts. Stable nontargeting controls and sensitivity analyses with alternative size factors are advisable.

6. **Guide-to-gene aggregation.** The supplied code reports guide-level inference. The GSE70038 analysis uses directional Stouffer aggregation only to make a transparent gene-level comparison; the Sanson benchmark uses the same rule. A production gene test should be benchmarked separately, ideally with control guides or phenotype permutations that preserve cross-guide dependence.

7. **Comparator assumptions.** MAGeCK-MLE offers information sharing and guide-efficiency estimation that the prototype lacks, but its normalized negative-binomial model and asymptotic Wald inference must also be diagnosed. Head-to-head results should be stratified by replication, phenotype type, signal strength, and composition shift rather than treating either method as a universal reference. Likewise, post-fit CNV correction should be accepted only after a null-gene diagnostic: it succeeds in A375 and over-corrects the longitudinal HT-29 screen.

14 Conclusion

CB^2-Reg extends the beta-binomial sampling principle behind a successful two-group method into a regression framework. This is an estimating-equation and design-matrix extension, not a claim that the default finite-sample estimator strictly contains the original CB^2 algorithm. Counts remain tied to their library depths, extra-binomial heterogeneity limits the leverage of deep sequencing, and uncertainty is determined by independent samples. What changes is the scientific interface: the target can now be a dose slope, time trend, adjusted treatment effect, interaction, or named contrast from an ordinary model matrix. RcppArmadillo kernels make that extension fast enough for genome-scale screening.

The evidence supports a focused, defensible claim. In the direct longitudinal use case, CB^2-Reg matches established methods in global ranking and gives the strongest high-precision recall. In the ordered-bin stress test, its continuous four-bin trend recovers substantially more validated biology than all-bin MAGeCK-MLE. After control calibration it retains a 22/26 versus 17/26 recovery advantage, has the best F1 and Matthews correlation among the independently rerun methods in the selected validation panel, and returns the non-targeting tail to its nominal rate. Backward-compatibility analyses recover established biology and show strong agreement with MAGeCK, while the A375 benchmark demonstrates competitive endpoint performance.

The same benchmarks prevent over-selling. MAGeCK-MLE is better calibrated in the reported simulation; Chronos captures mechanistic viability dynamics that a single slope omits; and Waterbear remains the appropriate generative model for correlated FACS partitions. MAUDE's 25/26 recovery comes with 406 calls and cannot be converted into a trustworthy precision or F1 estimate without complete labels. Copy-number correction is likewise a diagnostic choice, not an automatic improvement: it removes nuisance association in A375 and increases it in HT-29.

The resulting story is not "CB^2 versus a replacement." Original CB^2 answers the two-condition question; CB^2-Reg answers the coefficient question. Together they provide a coherent beta-binomial path from simple endpoint screens to quantitative and covariate-adjusted designs, with explicit boundaries that tell users when to move to a specialist model. The next improvements are therefore well defined: empirical-Bayes dispersion moderation, a native gene-level hierarchy, blocked or random-effect inference for repeated biological units, nonlinear time or dose effects, and covariance procedures that account for dispersion-estimation uncertainty.

References

[1] Keith A. Baggerly, Li Deng, Jeffrey S. Morris, and C. Marcelo Aldaz. Differential expression in SAGE: accounting for normal between-library variation. *Bioinformatics*, 19(12):1477–1483, 2003. doi: 10.1093/bioinformatics/btg173.

[2] Hyun-Hwan Jeong, Seon Young Kim, Maxime W. C. Rousseaux, Huda Y. Zoghbi, and Zhandong Liu. Beta-binomial modeling of CRISPR pooled screen data identifies target genes with greater sensitivity and fewer false negatives. *Genome Research*, 29(6):999–1008, 2019. doi: 10.1101/gr.245571.118.

[3] Keith A. Baggerly, Li Deng, Jeffrey S. Morris, and C. Marcelo Aldaz. Overdispersed logistic regression for SAGE: modelling multiple groups and covariates. *BMC Bioinformatics*, 5:144, 2004. doi: 10.1186/1471-2105-5-144.

[4] D. A. Williams. Extra-binomial variation in logistic linear models. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 31(2):144–148, 1982. doi: 10.2307/2347977.

[5] Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300, 1995. doi: 10.1111/j.2517-6161.1995.tb02031.x.

[6] Chad M. Toledo, Yu Ding, Pia Hoellerbauer, Ryan J. Davis, Ryan Basom, Elliott J. Girard, Eric Lee, Patricia Corrin, Traver Hart, Hamid Bolouri, Jared Davison, Qing Zhang, Jacqueline Hardcastle, Bruce J. Aronow, Christopher L. Plaisier, Nitin S. Baliga, Jason Moffat, Qi Lin, Xian Li, Dongjun Nam, Joo-Ha Lee, Steven M. Pollard, Jun Zhu, Jeffrey J. Delrow, Bruce E. Clurman, James M. Olson, and Patrick J. Paddison. Genome-wide CRISPR-Cas9 screens reveal loss of redundancy between PKMYT1 and WEE1 in glioblastoma stem-like cells. *Cell Reports*, 13(11):2425–2439, 2015. doi: 10.1016/j.celrep.2015.11.021.

[7] Beifang Wang, Meizhen Wang, Wenhu Zhang, Tengfei Xiao, Chen-Hao Chen, Aileen Wu, Fang Wu, Nicole Traugh, Xin Wang, Zibo Li, Shenglin Mei, Yan Cui, Shaoyang Shi, Jonathan J. Lipp, Michael Hinterndorfer, Johannes Zuber, Myles Brown, Wei Li, and X. Shirley Liu. Integrative analysis of pooled CRISPR genetic screens using MAGeCKFlute. *Nature Protocols*, 14(3): 756–780, 2019. doi: 10.1038/s41596-018-0113-7.

[8] Wei Li, Johannes Köster, Han Xu, Chen-Hao Chen, Tengfei Xiao, Jun S. Liu, Myles Brown, and X. Shirley Liu. Quality control, modeling, and visualization of CRISPR screens with MAGeCK-VISPR. *Genome Biology*, 16:281, 2015. doi: 10.1186/s13059-015-0843-6.

[9] Konstantinos Tzelepis, Hiroko Koike-Yusa, Etienne De Braekeleer, Yilong Li, Emmanouil Metzakopian, Oliver M. Dovey, Annalisa Mupo, Vera Grinkevich, Meng Li, Milena Mazan, Malgorzata Gozdecka, Shuhei Ohnishi, Jonathan Cooper, Miten Patel, Thomas McKerrell, Bin Chen, Ana Filipa Domingues, Paolo Gallipoli, Sarah Teichmann, Hannes Ponstingl, Ultan McDermott, Julio Saez-Rodriguez, Brian J. P. Huntly, Francesco Iorio, Cristina Pina, George S. Vassiliou, and Kosuke Yusa. A CRISPR dropout screen identifies genetic vulnerabilities and therapeutic targets in acute myeloid leukemia. *Cell Reports*, 17(4):1193–1205, 2016. doi: 10.1016/j.celrep.2016.09.079.

[10] Joshua M. Dempster, Isabella Boyle, Francisca Vazquez, David E. Root, Jesse S. Boehm, William C. Hahn, Aviad Tsherniak, and James M. McFarland. Chronos: a cell population dynamics model of CRISPR experiments that improves inference of gene fitness effects. *Genome Biology*, 22:343, 2021. doi: 10.1186/s13059-021-02540-7.

[11] Joshua M. Dempster. Chronos: Gene fitness effect inference from CRISPR. Figshare dataset, 2021.

[12] Broad DepMap. DepMap Public 20Q2. Figshare dataset, 2020.

[13] Harold Pimentel, Jacob W. Freimer, Maya M. Arce, Christian M. Garrido, Alexander Marson, and Jonathan K. Pritchard. A model for accurate quantification of CRISPR effects in pooled FACS screens. *bioRxiv*, 2024. doi: 10.1101/2024.06.17.599448.

[14] Carl G. de Boer, John P. Ray, Nir Hacohen, and Aviv Regev. MAUDE: inferring expression changes in sorting-based CRISPR screens. *Genome Biology*, 21:134, 2020. doi: 10.1186/s13059-020-02046-8.

[15] Kendall R. Sanson, Ruth E. Hanna, Mudra Hegde, Kathleen F. Donovan, Curtis Strand, Meagan E. Sullender, Emma W. Vaimberg, Alex Goodale, David E. Root, Federica Piccioni, and John G. Doench. Optimized libraries for CRISPR-Cas9 genetic screens with multiple modalities. *Nature Communications*, 9:5416, 2018. doi: 10.1038/s41467-018-07901-8.

[16] Broad DepMap. DepMap Public 19Q3. Figshare dataset, 2019.

[17] Robin M. Meyers, Jordan G. Bryan, James M. McFarland, Barbara A. Weir, Ann E. Sizemore, Han Xu, Neekesh V. Dharia, Phillip G. Montgomery, Glenn S. Cowley, Stacey Pantel, Alex Goodale, Yoon Lee, Levi D. Ali, Guozhi Jiang, Robi Lubonja, William F. Harrington, Matthew Strickland, Ting Wu, David C. Hawes, Victor A. Zhivich, Matthew R. Wyatt, Zahra Kalani, Jennifer J. Chang, Michael Okamoto, Kimberly Stegmaier, Todd R. Golub, Jesse S. Boehm, Francisca Vazquez, David E. Root, William C. Hahn, and Aviad Tsherniak. Computational correction of copy number effect improves specificity of CRISPR-Cas9 essentiality screens in cancer cells. *Nature Genetics*, 49(12):1779–1784, 2017. doi: 10.1038/ng.3984.

[18] Wei Li, Han Xu, Tengfei Xiao, Le Cong, Michael I. Love, Feng Zhang, Rafael A. Irizarry, Jun S. Liu, Myles Brown, and X. Shirley Liu. MAGeCK enables robust identification of essential genes from genome-scale CRISPR/Cas9 knockout screens. *Genome Biology*, 15(12):554, 2014. doi: 10.1186/s13059-014-0554-4.